Work Note v3 — Analysis of the Gas-Flow Condition Data

To: Prof. Taniike
Follows: Work Note v2 and your reply with `OCM_data.csv`. This note reports only what is new.
Code: `phase11_condition_baseline.py`, `phase12_runs_and_coverage.py`, `phase8b_label_sweep.py`, `phase13_condition_value.py`, `phase14_crossprep_pft.py`. Every number below is stored in the JSON file of the same name.

Protocol. As in v2: - All measurements of a catalyst stay together, in either training or testing. - Five repeats use different random splits. - A difference between two models is reported with a 95% confidence interval (CI), from resampling catalysts. - Yields and errors are in percentage points of C_2 yield ("points").

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1. What the condition data shows

- The design has 5 temperatures (700–900 °C) and 27 gas-flow conditions per temperature.
- The 27 conditions form a full 3 × 3 × 3 design:
  - total flow of 10, 15 or 20 (the file gives no unit)
  - Ar fraction of 0.70, 0.40 or 0.15
  - CH_4:O_2 ratio of 2, 4 or 6
- The condition file answers both questions in v2 (§8). Your mass-balance explanation accounts for the incomplete groups.
- The readings at one temperature are therefore 27 different conditions. v2 (§5) called them near-identical readings, which was wrong. The training label is unchanged. It is now understood as a median over up to 27 different conditions, not over repeated readings.

2. Prediction of individual measurements

- Adding the three gas-flow columns lowers the RMSE for unseen catalysts from 2.90 to 2.66 points.
- The reduction is 0.24 points (8.1%), with a 95% CI of 0.22 to 0.26.
- Most of the gain comes from predicting how yield changes between conditions at the same temperature.
- The model settings were tuned after v2, which is why the baseline here differs from v2's 2.94.

3. Ranking of catalysts

The conditions do not improve the ranking of catalysts by best yield.

- **The comparison.** Each catalyst is scored on the same 18 conditions: CH_4:O_2 = 2 at 800 and 850 °C. 465 of the 917 catalysts were measured at all 18. We chose this set by measurement coverage alone, before comparing models.
- **The metric.** The rank correlation between predicted and observed best yield over those 18 conditions: 0.732 with the conditions and 0.727 without.
- **The result.** The difference is +0.005 (95% CI −0.007 to +0.015), which is no detectable change.
- **Two other scorings.**
  - Scoring each catalyst only over the conditions that passed the mass-balance check shows an apparent gain. It disappears on the 771 catalysts with comparable measurement effort (the equal-effort set of v2), so we do not count it as a gain.
  - Taking the best prediction over all 135 conditions ranks slightly worse (−0.014). That comparison is biased, because excluded runs are missing from the observed best yield.

4. Where the conditions add value

Choosing conditions for a new catalyst. The condition-aware model of §2 ranks a new catalyst's 135 conditions (5 temperatures × 27).

Rule for choosing three conditions to run	Mean shortfall from the catalyst's best yield
Model ranking	1.75 points
Best three conditions for the same support oxide, taken from other catalysts	1.90 points

- The model's advantage is 0.15 points (95% CI 0.07 to 0.23), and it holds in all five repeats.
- The mean best yield is 10.4 points, so the advantage is 1.4% of it. The effect is consistent but small.
- About 16% of chosen conditions were never measured under either rule; these count as failed runs. Skipping them instead gives the same advantage.

A reduced measurement set. From training catalysts only, we chose the 10 of the 135 conditions that most often give a catalyst its best yield.

- **What was chosen.** All 10 have CH_4:O_2 = 2, and most have total flow 20. The set varies slightly between splits: 7 conditions were chosen in all 25 splits.
- **Result.** On held-out catalysts, the best yield over these 10 conditions ranks catalysts with a rank correlation of 0.969 against the full data. This is the mean over five repeats; 92% of catalysts had at least one of the 10 measured.
- **Comparison.** Random sets of 10 conditions give 0.917 on average (97.5th percentile 0.925). A second selection rule, which adds the conditions that most improve the ranking, gave 0.939.
- **Caution.** The result is optimistic. 84.5% of catalysts have their best yield at CH_4:O_2 = 2, and the full-data best yield comes from the same runs. A single fixed set is needed before any prospective test.
- **Training on the reduced labels.** A composition model trained on the best yield over the second rule's 10 conditions ranked 0.009 lower (95% CI −0.018 to +0.001). A small loss is possible.

Per-catalyst response models. Two models of each catalyst's response to the conditions were less accurate than the model of §2, so we have stopped this work.

5. Training label

We keep the per-temperature median, then the highest of these medians, as in v2.

- **Stability.** Removing each catalyst's single highest reading moves the median label by 0.01 points. It moves the maximum by 0.38 points. Both are medians over catalysts.
- **New comparison.** The upper quartile ranks slightly better than the maximum (+0.006, 95% CI +0.001 to +0.012), but not better than the median (+0.001, 95% CI −0.002 to +0.005). We therefore keep the median.
- **Caution.** 84.5% of catalysts have their best yield at CH_4:O_2 = 2, the lowest ratio tested. The true optimum for these catalysts may lie outside the tested range.

6. Literature data across preparation methods

In v2, one model trained on laboratory and impregnation literature data together (a "merge") ranked non-impregnation literature better than our two-stage model: rank correlation 0.388 versus 0.318.

- **Two-stage variants.**
  - We tested three further variants. In one, every stage-1 prediction comes from a model that did not see that row.
  - None of these is the original two-stage model: none uses the DRST filter, and stage 1 sees literature data only.
  - None ranked better than the merge. Two ranked lower, and one showed no detectable difference.
- **Rescaling.**
  - One element of our two-stage method helps on its own: rescaling the literature yields onto the laboratory yield distribution (quantile normalisation) before merging.
  - Rank correlation rises from 0.388 to 0.422. The difference is +0.034 (95% CI +0.018 to +0.050). It holds in all five repeats and under the median label.
  - Enrichment of top performers shows no detectable change.
- **Why this differs from v2.** In v2 (§1), the same rescaling made no measurable difference for laboratory catalysts. It helps here, where the literature yields differ in scale from the laboratory yields.
- **Status.** This comparison was added during the analysis, so we treat it as exploratory.

- **Scope.** The literature file has no condition columns, so the gas-flow data cannot enter this part.

7. Two observations on the data

Catalysts with two sets of runs. - For 28 catalysts, 1,906 conditions appear twice. - The two readings of a condition differ by a median factor of 2.3. - The higher readings form a smooth pattern across conditions, and so do the lower readings. A random mix of the two does not. This suggests two runs at different yield levels. - 10 of the 92 catalysts in the top 10% by best yield are among these 28; chance would give 2.8. If the two readings are averaged, 4 remain.

Excluded runs. - The rank correlation between a catalyst's number of missing conditions (out of 135) and its best yield is −0.29. Exclusions are therefore more frequent for weaker catalysts. - Within a catalyst, excluded conditions rank slightly lower than measured ones, by 2–3 percentile points. The difference is detectable, but it is below the 5-point threshold we set in advance. - Only 46% of excluded conditions could be tested this way, because the test needs the same condition measured at a neighbouring temperature.

8. Questions and next steps

Both questions concern records that are not in the data file.

1. **For the 28 catalysts with two sets of runs, is there a record of the run date or catalyst batch?** About 6 of the 92 top-10% places depend on how the two sets are treated.
2. **Are the records excluded for poor mass balance still available?** - Our indirect test reached only 46% of them. - They would show whether each catalyst's best observed yield is complete. - They would also show how a shorter protocol should treat a failed run.

Next steps. - Build on the rescaling result (§6). - We will add the DRST filter and an out-of-fold stage-1 feature, one at a time, with the rule for success fixed in advance. - Stage 1 will not be trained on laboratory rows, the source of the leakage reported in v2. - Fix a single 10-condition set (§4) and test how it ranks catalysts not used to choose it, before any prospective use.